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Program ISC
Specialization Software Engineering

Integrating GitLab with Moodle

Context

Learning Management Systems (LMS) such as Moodle play a central role in organizing higher education courses. At the same time, in software development, Version Control Systems (VCS) such as GitLab have become essential tools for source code management, project coordination, and collaboration. In computer science education, these systems are also widely used for distributing assignments and evaluating practical work. However, despite their complementary roles, LMS and VCS platforms typically operate independently, resulting in fragmented workflows. The need for a unified solution has become even more apparent following the announcement of the discontinuation of GitHub Classroom on 28 August 2026.

Objectives

This project aims to be an open-source, extensible Moodle plugin that combines LMS-based assignment management with Git-based development workflows. The system should automates repetitive administrative tasks, including repository provisioning, group management, synchronization, solution publication, and grading processes. It should reduce manual overhead, while providing students with a centralized environment for accessing assignments, source code, instructions, and feedback.

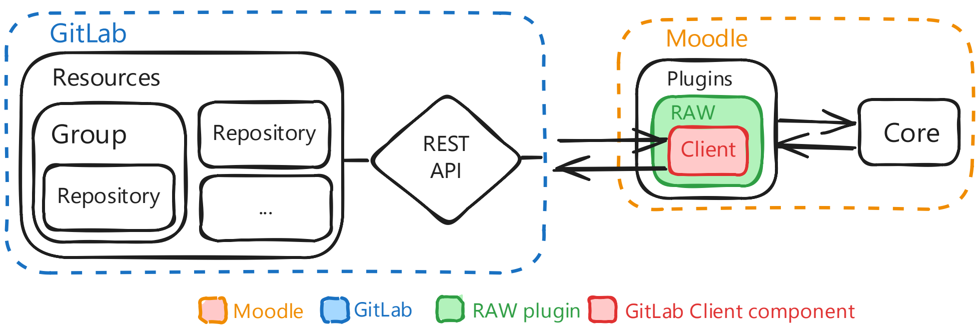

Figure 1: RAW plugin interactions with GitLab

Results

This project addresses the lack of integration between Learning Management Systems and version control systems in computer science education by introducing **Repository Assignment Workflow** (**RAW**).

RAW is built around three key principles: LMS-native integration to reduce platform fragmentation, flexible automation instead of rigid workflows, and a modular open-source architecture that supports extensibility. The evaluated existing solutions do not provide an equivalent combination of these characteristics, as they often lack native LMS integration, rely on proprietary ecosystems, or require significant manual intervention.

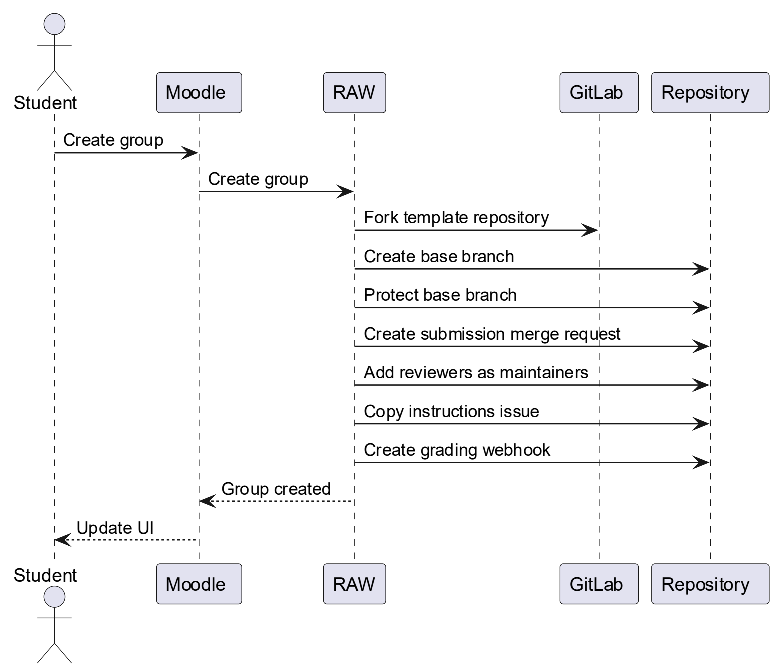

Figure 2: Student group creation flow

Conclusion

Beyond improving day-to-day assignment management, **RAW** establishes a defined workflow that bridges educational platforms and software development tools. Its modular architecture and open-source design enable future extensions and adaptations. The direct integration into the LMS provides a unified and centralized platform that offers a more intuitive user experience while seamlessly integrating with the institution's existing educational ecosystem. This approach reduces platform fragmentation and standardizes practical assignment workflows. **RAW** therefore provides an efficient, consistent, and scalable educational experience for both instructors and students.

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